

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Answer:

Find mean, standard deviation and standard error in R.

```

1 mean_y <- mean(y)
2 s_y <- sd(y)
3 se <- s_y / sqrt(length(y))
4 # Mean = 98.44
5 # Standard Deviation = 13.0929
6 # Standard Error = 2.6186

```

Sample size is less than 30, use t-distribution.

```

1 t_score <- qt(0.95, df = length(y) - 1)
2 # t_score = 1.711

```

In above function $qt()$, the argument $p = 1 - 0.1/2 = 0.95$

```

1 ci_lower <- mean_y - t_score*se
2 ci_upper <- mean_y + t_score*se
3 # 90% Confident Interval is (93.9599, 102.9201)

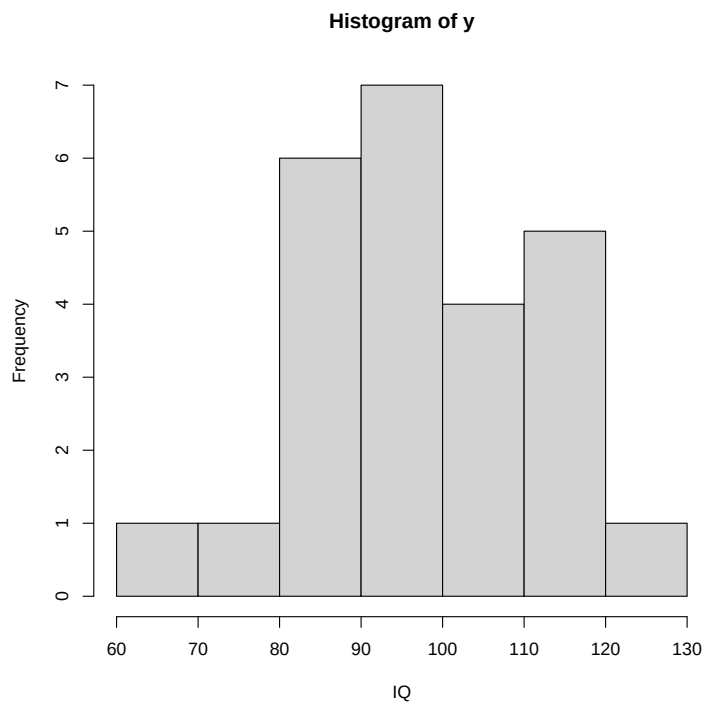
```

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Answer:

Assumption: data is randomly selected and the population is normally distributed.



Hypothesis:

$$H_0 : \mu \leq 100$$

$$H_A : \mu > 100$$

Calculate test statistics:

Since the sample size is small we use t-score to calculate test statistic:

$$t = \frac{\bar{y} - \mu_0}{\sigma_{\bar{y}}}$$

We can find t in R.

```

1 t <- (mean_y - 100) / se
2 # t = -0.596

```

Calculate P-value in R:

```
1 p_value <- pt(q=t, df = length(y)-1, lower.tail = FALSE)
2 # p_value = 0.7215
```

Conclusion:

Since $P\text{-value} > \alpha(0.05)$, we fail to reject the null hypothesis. There is insufficient evidence to conclude that the average student IQ in this school is greater than the average IQ score of 100 among all schools in the country.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

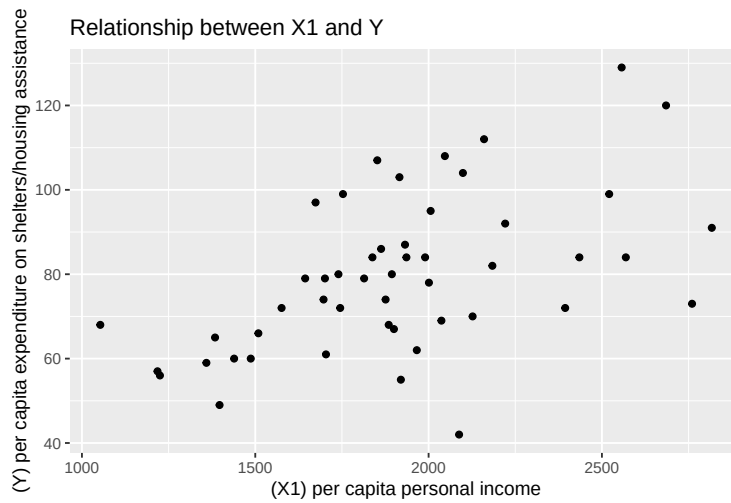
```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

- Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?

Answer:

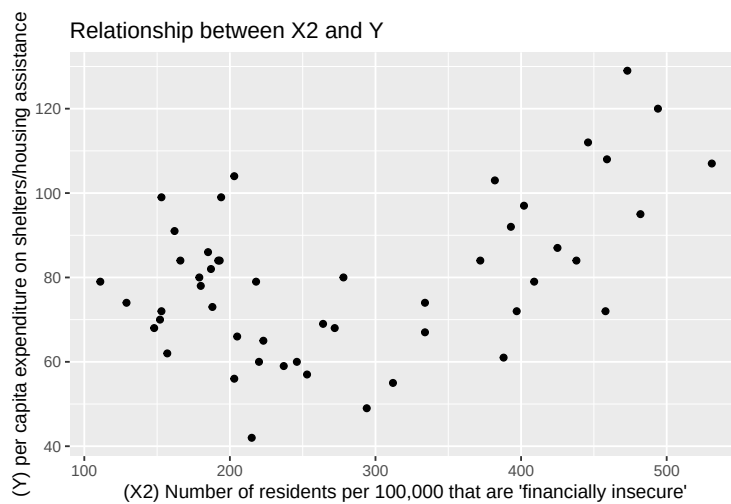
First, plot scatter plot to show the relationship between Y (per capita expenditure on shelters/housing assistance in state) and $X1$ (per capita personal income in state).

```
1 # plot relationship between y and x1 using scatter plot
2 x1_y_scatter <- ggplot(expenditure, aes(x=X1, y=Y)) +
3   geom_point() +
4   labs(
5     title = "Relationship between X1 and Y",
6     x = "(X1) per capita personal income",
7     y = "(Y) per capita expenditure on shelters/housing assistance"
8   )
9 ggsave("x1_y_scatter.pdf", x1_y_scatter, width = 6, height = 4)
```

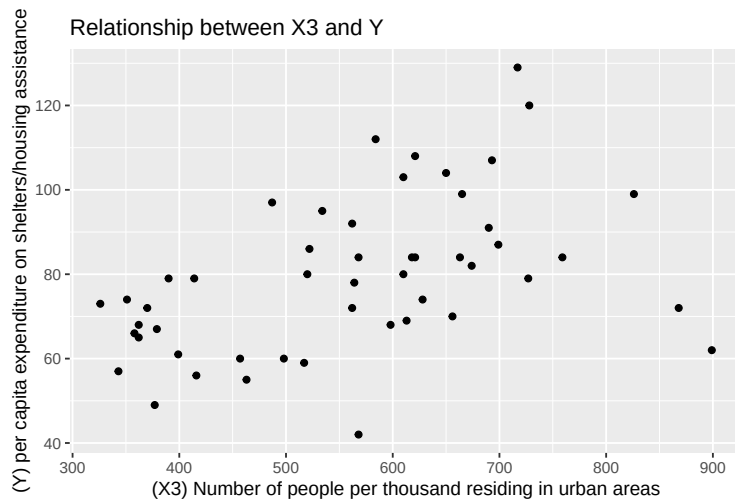


From this graph, I can see there is a **positive, strong** and **linear** association between $X1$ and Y . When $X1$ increases, Y increases.

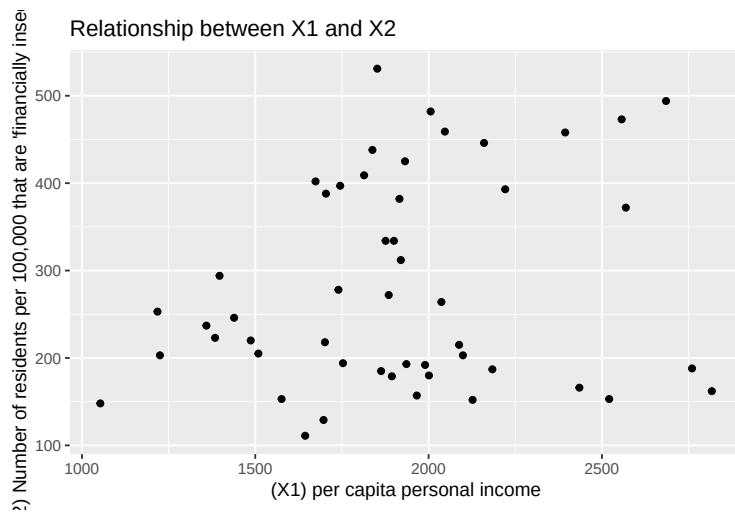
Similarly, I can plot scatter graph for $X2$ and Y , $X3$ and Y , $X1$ and $X2$, $X1$ and $X3$, $X2$ and $X3$.



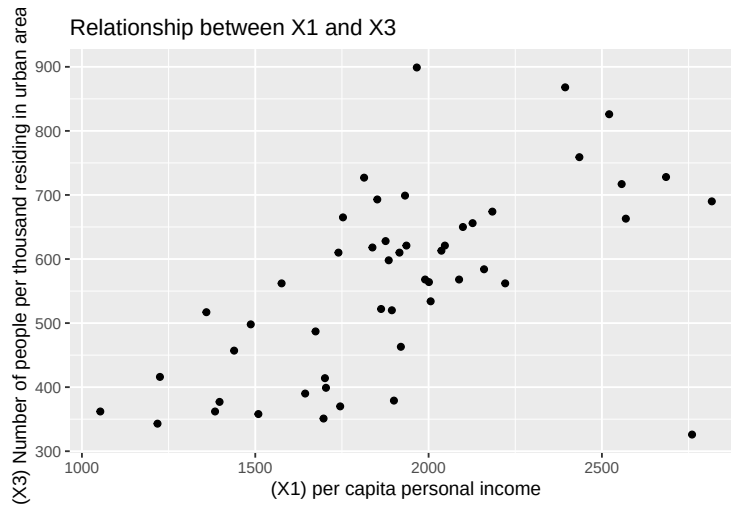
From this graph, there is a **nonlinear** association between $X2$ and Y .



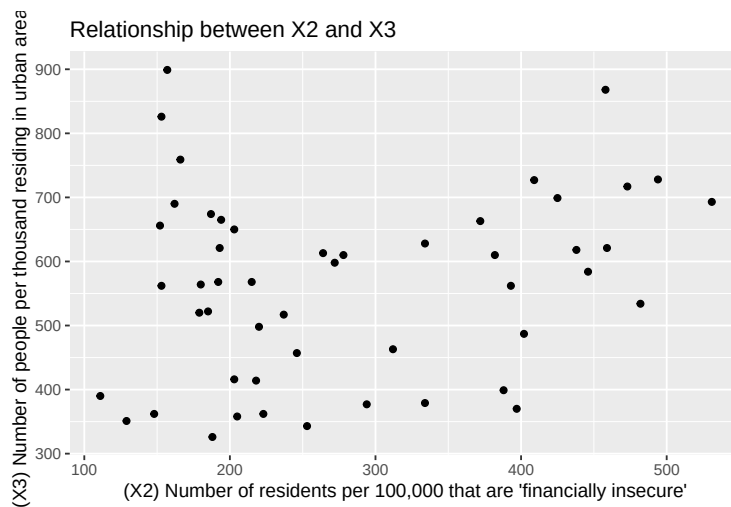
From this graph, there is a **positive**, **strong** and **linear** association between X_3 and Y , with some outliers when X_3 (the number of people per thousand residing in urban areas) is around 900.



From this graph, there is a **positive**, **weak** and **linear** association between X_1 and X_2 .



From this graph, there is a **positive**, **strong** and **linear** association between $X1$ and $X3$, with one outlier when $X1$ (per capita personal income) is around 2750



From this graph, there is **nonlinear** correlation between $X2$ and $X3$.

- Please plot the relationship between Y and $Region$? On average, which region has the highest per capita expenditure on housing assistance?

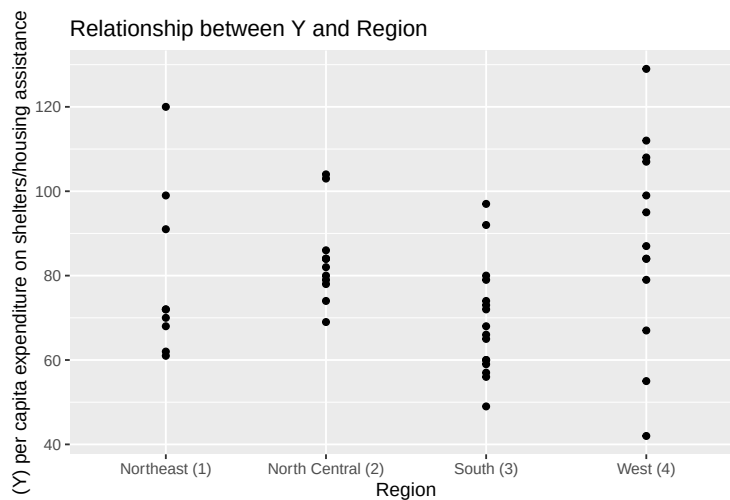
Answer:

To make the number from Region easier to be understood. I can change Region to factor with label names of Northeast, North Central, South and West.

```
1 expenditure$Region <- factor(expenditure$Region ,
2   levels = c(1, 2, 3, 4) ,
3   labels = c("Northeast (1)", "North Central (2)", "South (3)", "West (4)
4     ")))
```

Then create scatter plot.

```
1 # relationship between Y and Region
2 y_region_scatter <- ggplot(expenditure , aes(x=Region , y=Y)) +
3   geom_point() +
4   labs(
5     title = "Relationship between Y and Region" ,
6     x = "Region" ,
7     y = "(Y) per capita expenditure on shelters/housing assistance"
8   )
9 ggsave("y_region_scatter.pdf" , y_region_scatter , width = 6, height = 4)
```

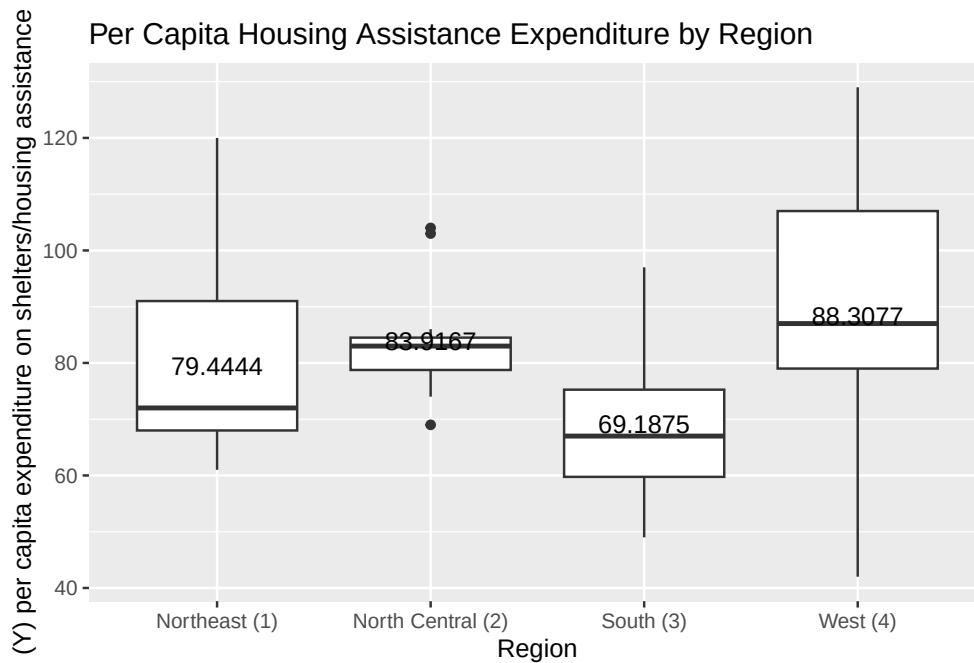


To find the average, I can use box plot to compare the mean per capita expenditure among different region.

```

1 # Use Region as factor to plot box plot.
2 y_region_box <- ggplot(expenditure, aes(x = Region, y = Y)) +
3   geom_boxplot() +
4   stat_summary(
5     fun = mean,
6     geom = "text",
7     aes(label = after_stat(round(y,4))),
8   ) +
9   labs(
10    title = "Per Capita Housing Assistance Expenditure by Region",
11    x = "Region",
12    y = "(Y) per capita expenditure on shelters/housing assistance"
13  )
14
15 ggsave("y_region_box.pdf", y_region_box, width = 6, height = 4)

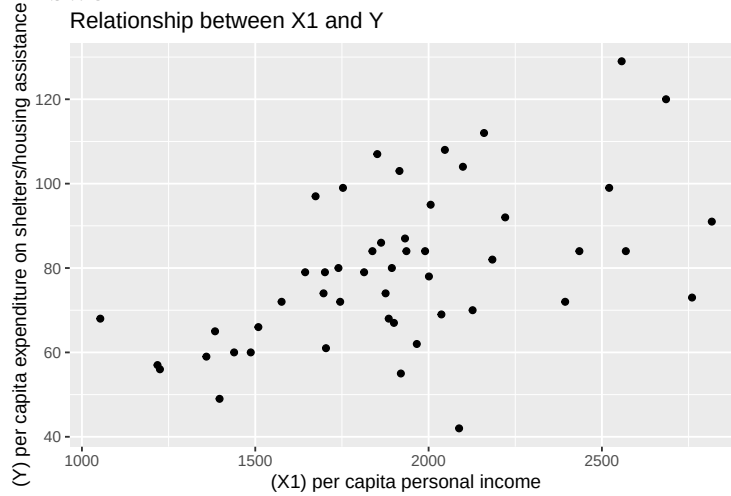
```



On average, US states that are categorized as 4(West) has the highest per capita expenditure (88.3077) on housing assistance.

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

Answer:

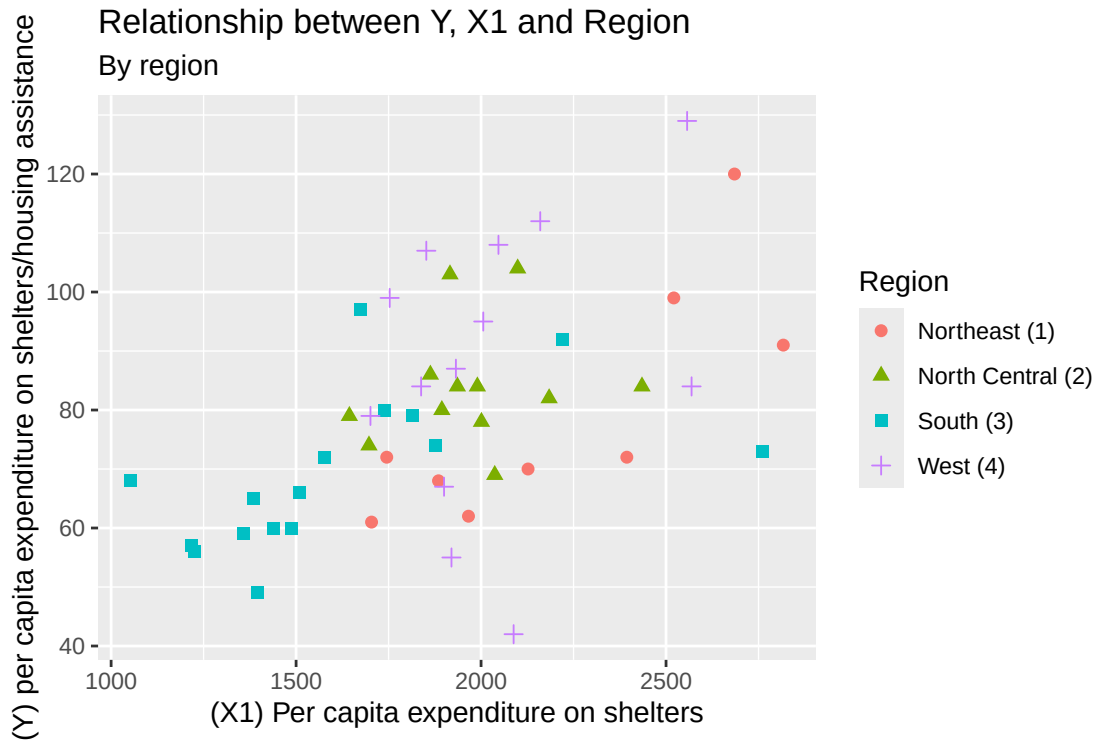


In this graph, Y and $X1$ have a strong, positive and linear relationship, as Y increases, $X1$ increases as well.

```

1 # re-creating above relationship between Y and X1, including one more
  variable Region
2 y_x1_region <- ggplot(expenditure, aes(x=X1, y=Y,
3                                         colour = Region, shape = Region)) +
4   geom_point(size = 2) +
5   labs(
6     title = "Relationship between Y, X1 and Region",
7     x = "(X1) Per capita expenditure on shelters",
8     y = "(Y) per capita expenditure on shelters/housing assistance",
9     subtitle = "By region",
10  )

```



South region is clustered mostly in the low Y and low X1, Northeast region is clustered in the high Y and high X1. The North Central and West regions are clustered in the middle Y and X1 area.